# WorkSheet 1: Counting

*Exercises marked ⋆ should be done in priority. Exercises marked (lecture) will be solved in the lecture, and those marked (homework) are left as homework.*

**Exercise 1** (⋆)**.** How many possible initials (consisting of 2 letters) can a person have? How many students are needed in a class to guarantee that at least two of them share the same initials?

**Exercise 2** (⋆)**.** How many 3-letter words can be formed with no restrictions (letters may be repeated; the vowels are A, E, I, O, U and the other 21 letters are consonants)? How many can be formed with at least 2 consonants? With exactly 2 consonants and 1 vowel? With at most 1 vowel?

**Exercise 3** (⋆)**.** Two consecutive experiments are conducted. The first experiment has $m$ possible outcomes. For the $i^{th}$ outcome of the first experiment, the second experiment has $n_i$ possible outcomes, where $i = 1, \cdots, m$. What is the total number of possible outcomes for both experiments combined?

**Exercise 4** (⋆)**.** The postman arrives with 5 (different) letters in a building where there are 7 mailboxes. Determine the number of ways to place the 5 letters in the 7 mailboxes if: i. a mailbox can contain all 5 letters. ii. a mailbox can only contain one letter.

**Exercise 5** (⋆)**.** Consider $\mathbb{E} = \{1, 2, \cdots, p\}$ and $\mathbb{F} = \{1, 2, \cdots, q\}$. Determine the number of

1. Maps from $\mathbb{E}$ to $\mathbb{F}$?
2. Injections from $\mathbb{E}$ to $\mathbb{F}$?
3. Bijections from $\mathbb{E}$ to $\mathbb{F}$?

**Exercise 6** (⋆)**.** From a committee of 10 members, $\{A, B, C, D, E, F, G, H, I, J\}$, a chairperson, a secretary, and a treasurer are to be chosen.

1. Knowing that accumulation is not allowed (a person cannot hold more than one office), in how many ways can these offices be allocated if
   (a) There are no restrictions?
   (b) A and B cannot both hold an office?
   (c) C and D both hold an office or neither does?
   (d) F must have a task?
   (e) F only accepts the task of chairperson?
2. Knowing that accumulation is allowed (a person may hold several offices), in how many ways can these offices be allocated if there are no restrictions?

**Exercise 7.** At a particular company, employees have a seven-digit security code. Five of the digits must come from $\{1, 4, 5, 7\}$ with repetition allowed. The remaining two digits must come from $\{2, 3, 9\}$ with no repetition.

1. How many security codes are possible?
2. How many of these begin with 1?
3. How many of these end with 5?
4. How many of these neither begin with 1 nor end with 5?
5. Suppose that this company has 150,000 employees. Can we be sure that two employees have the same security code?

**Exercise 8** ($\star$)**.** The 10 volumes of an encyclopedia are arranged randomly on a shelf. In how many ways can they be arranged if there are no restrictions? How many ways are there if volumes 7, 6, and 5 must be placed next to each other in that specific order?

**Exercise 9.** Four Algerians, three Libyans and two Tunisians must be lined up in a race. Athletes of the same nationality must stay together. How many arrangements can we imagine?

**Exercise 10** ($\star$)**.** $m$ boys and $n$ girls are to be arranged, where $m, n \in \mathbb{N}$. Find the number of ways this can be done in each of the following cases.

I) In a row:

(a) There are no restrictions;

(b) No boys are adjacent ($m \leq n+1$);

(c) The $n$ girls form a single block;

(d) A particular boy and a particular girl must be adjacent.

II) Around a round table (two arrangements that differ only by a rotation are the same), redo a) and c).

**Exercise 11.** In how many ways can 8 people be seated

1. In a row, if:

(a) there are no restrictions?

(b) two persons A and B sit next to each other?

(c) there are 4 adults and 4 children, and each adult has only children as neighbors and vice versa?

(d) the number of adults is 6, they must stay together?

(e) there are 4 children, each accompanied by his mother who stays next to him?

2. Around a round table (rotations are not distinguished), redo a) and c).

**Exercise 12.** How many football teams (of 11 players) can be formed from 22 people without restrictions? How many teams can be formed if among the 22 people there are 3 goalkeepers and exactly one of them must be included?

**Exercise 13** ($\star$)**.** A company consists of 2 managers, 3 administrators and 11 workers. Four people are invited to an event. How many ways can they be chosen if:

(i) There are no restrictions?

(ii) There is 1 manager, 2 administrators, and 1 worker?

(iii) Workers and administrators cannot be together (the four people cannot include both a worker and an administrator)?

**Exercise 14** ($\star$)**.** In the exam, the student has to answer 5 out of 8 questions. How many choices does he have? How many choices would he have if he had to answer at least 3 out of the first 5 questions?

**Exercise 15.** On Eid, you plan to visit 7 family members (the order of the visits matters). In how many ways can you visit all of them if all visits are made on the same day? How many ways are there if you visit 4 of them on Eid and the remaining 3 the next day?

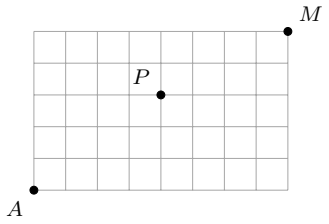
**Exercise 16.** Consider a population of N individuals. We take $n$ successive samples from this population, with replacement. How many sequences of length $n$ exist in which individual **A** is taken:

1. exactly $k$ times?
2. exactly $k$ times in the first $r$ draws?
3. for the $k^{th}$ time in the $r^{th}$ draw ?

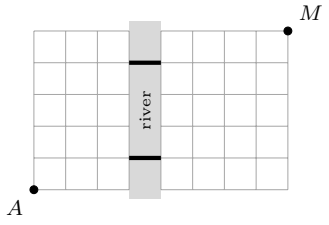
**Exercise 17** ($\star$)**.**
A city grid is modeled as a 2D Cartesian plane. A courier must travel from the Central Station $A(0,0)$ to the Delivery Hub $M(8,5)$, moving only one unit to the Right (R) or one unit Up (U) at a time. Calculate the number of paths from $A$ to $M$ under the following independent conditions:

1. There are no additional restrictions.
2. The path passes through the intersection $P(4,3)$.
3. A street parade is blocking $P(4,3)$: the courier must **avoid** $P(4,3)$.
4. **The river constraint:** a river runs between the lines $x=3$ and $x=4$, and it can only be crossed by Bridge 1, from $(3,1)$ to $(4,1)$, or by Bridge 2, from $(3,4)$ to $(4,4)$.

Questions 1–3




Question 4




**Exercise 18** ($\star$)**.** There are $n$ sticks lined up in a row, and $k$ of them are to be chosen.

1. How many choices are there?
2. How many choices are there if no two of the chosen sticks can be consecutive?
3. How many choices are there if there must be at least $l$ sticks between each pair of chosen sticks?

**Exercise 19.** Find the number of ways of forming a group of $2k$ people from $n$ couples, where $k, n \in \mathbb{N}$ with $2k \leq n$, in each of the following cases:

1. There are exactly $k$ couples in such a group (the group consists of $k$ complete couples);

2. No couples are included in such a group;
3. At least one couple is included in such a group;
4. Exactly two couples are included in such a group.

**Exercise 20** (lecture)**.** Prove in two different ways the following identity (known as the absorption/extraction identity), for $1 \leq k \leq n$:

$$k\binom{n}{k} = n\binom{n-1}{k-1}$$

**Exercise 21** (lecture)**.** Establish the "committee-subcommittee" identity, for $0 \leq r \leq k \leq n$:

$$\binom{n}{k}\binom{k}{r} = \binom{n}{r}\binom{n-r}{k-r}$$

**Exercise 22** ($\star$)**.**
1. For integers $n \geq k \geq 2$, give a combinatorial proof of the identity

$$k(k-1)\binom{n}{k} = n(n-1)\binom{n-2}{k-2}.$$

2. For $n \geq 0$, compute the sum

$$S = \binom{n}{0} + 2\binom{n}{1} + 3\binom{n}{2} + \cdots + (n+1)\binom{n}{n}.$$

**Exercise 23** ($\star$)**.** Prove the following identities, which are extensions of Pascal's rule:

1. $\binom{n}{k} = \binom{n-2}{k-2} + 2\binom{n-2}{k-1} + \binom{n-2}{k}$
2. $\binom{n}{k} = \binom{n-3}{k-3} + 3\binom{n-3}{k-2} + 3\binom{n-3}{k-1} + \binom{n-3}{k}$
3. Generalize these formulas to an identity relating $\binom{n}{k}$ to a sum of terms involving $\binom{n-m}{j}$ for a general $m \leq n$.

**Exercise 24** ($\star$)**.** Show the "hockey-stick" identity:

$$\binom{n+1}{k+1} = \sum_{j=0}^{n-k} \binom{n-j}{k} = \binom{k}{k} + \binom{k+1}{k} + \cdots + \binom{n}{k}$$

Deduce from this identity the formulas for:

1. $\displaystyle\sum_{i=1}^{n} i$
2. $\displaystyle\sum_{i=1}^{n} i(i+1)$

3. $\sum_{i=1}^{n} i(i+1)(i+2)$
4. $\sum_{i=1}^{n} i^2$
5. $\sum_{i=1}^{n} i^3$

**Exercise 25** ($\star$)**.** Give a combinatorial proof of the following identity:

$$\sum_{k=0}^{n} k\binom{n}{k} = n2^{n-1}$$

**Exercise 26.** Calculate the following sums:

1. $S_1 = \binom{n}{0} + \binom{n}{2} + \binom{n}{4} + \cdots$
2. $S_2 = \binom{n}{1} + \binom{n}{3} + \binom{n}{5} + \cdots$

**Exercise 27** ($\star$)**.** Calculate the sums $\sum_{k=0}^{n} \frac{1}{k+1}\binom{n}{k}$ and $\sum_{k=0}^{n} \frac{2^k}{k+1}\binom{n}{k}$.

**Exercise 28** ($\star$)**.** Establish the following identity, for $r \leq n$:

$$\sum_{k=r}^{n} \binom{n}{k}\binom{k}{r} = \binom{n}{r}2^{n-r}$$

**Exercise 29.**

1. Give a combinatorial proof that:

$$\sum_{k=1}^{n} k\binom{n}{k}^2 = n\binom{2n-1}{n-1}$$

2. Show analytically that $n\binom{2n-1}{n-1} = \frac{n}{2}\binom{2n}{n}$.

**Exercise 30.** Prove that:

$$\sum_{k=0}^{2n} (-1)^k \binom{2n}{k}^2 = (-1)^n \binom{2n}{n}$$

**Exercise 31.**

- Establish the following identity, for $r < n$:

$$\sum_{k=r}^{n} \binom{n}{k}\binom{k}{r}(-1)^{n-k} = 0$$

- Show that for $1 \leq k \leq n-1$:

$$\binom{n}{k}^2 > \binom{n}{k-1}\binom{n}{k+1}$$

- Calculate the product:
$$\prod_{k=0}^{n} \binom{n}{k} = \binom{n}{0}\binom{n}{1}\cdots\binom{n}{n}$$

**Exercise 32.** Calculate the following sums:

1. $S_1 = \binom{n}{0} + \binom{n}{3} + \binom{n}{6} + \cdots$
2. $S_2 = \binom{n}{1} + \binom{n}{4} + \binom{n}{7} + \cdots$
3. $S_3 = \binom{n}{2} + \binom{n}{5} + \binom{n}{8} + \cdots$

(Hint: Consider $(1+x)^n$ where $x$ is a complex cube root of unity, e.g., $x = e^{i2\pi/3}$.)

**Exercise 33.** A decorative display consists of a sequence of $n \geq 1$ LED lights; each light is given one colour. Find the number of possible colourings of the sequence in each case:

1. The colours are {Red, Blue, Green, Black}, with no restrictions.
2. Same colours; the number of Red lights is **even** and the number of Blue lights is **odd**.
3. The colours are {Red, Blue, Green, Black, White, Yellow}; the total number of lights that are White or Yellow is **even**.
4. The colours are {Red, Blue, Green}; the number of Red lights and the number of Blue lights have the **same parity**.
5. The colours are {Red, Blue, Green, Black}; the numbers of Red and of Blue lights are both even, and there is **at least one** Green light.

(Hint: expand $(r + b + g + \cdots)^n$ with the multinomial theorem, then give the values $\pm 1$ to some of the variables.)

**Exercise 34.** Using the binomial expansion, give an approximate value to $10^{-3}$ (i.e., 3 decimal places) for the following:

1. $(0.98)^{10}$
2. $(1.001)^{100}$
3. $(\frac{899}{900})^{50}$

**Exercise 35** ($\star$)**.** Prove Vandermonde's Identity, first using a combinatorial argument and then analytically (e.g., by considering the coefficient of $x^r$ in $(1+x)^n(1+x)^m$):
$$\binom{n+m}{r} = \sum_{k=0}^{r} \binom{m}{k}\binom{n}{r-k}$$
Deduce from this identity that $\displaystyle\sum_{k=0}^{n} \binom{n}{k}^2 = \binom{2n}{n}$.

**Exercise 36** ($\star$)**.** How many different words can be formed using all the letters of each of the words: ALGER, ANNABA, TIZIOUZOU, MISSISSIPPI?

**Exercise 37** ($\star$)**.** At a Blood Donation Center, there are $n$ people standing in a line waiting to give blood. Each person has one of the 4 blood types A, B, AB or O. Find the number of possible sequences of blood types for the line in each case below:

1. There are no restrictions.
2. No two people standing next to each other have the same blood type.
3. Exactly $k$ people have type O.
4. The line consists *only* of types A and B, and no two people of type A are adjacent.
5. The line contains exactly $n_1$ people of type A, $n_2$ of type B, $n_3$ of type AB and $n_4$ of type O ($n_1 + n_2 + n_3 + n_4 = n$).
6. There is at least one person of type O.

**Exercise 38** ($\star$)**.**

1. Expand $(2x^3 - 3y^2)^5$ using the binomial theorem.
2. Expand $(x^2 - 2y + 4)^3$ using the multinomial theorem.
3. Give the coefficients of the monomials $x^{14}y^5z^3$ and $xy^2z^3$ in the expansion of $(2x^2 - 3y + z)^{15}$.

**Exercise 39.** What is the coefficient of $x^{30}y^{20}z^{50}$ in the expansion of $(x - 2y + 3z - \frac{1}{2}w)^{100}$?

**Exercise 40** ($\star$)**.** How many pieces are there in a domino game, knowing that on each piece there are two symbols chosen among: $\{\text{blank}, 1, 2, 3, 4, 5, 6\}$? $\{\text{blank}, 1, 2, \ldots, 9\}$? (A piece is determined by its symbols, not by their order, and the same symbol may appear twice.) Generalize to triominoes and tetraominoes (3 and 4 symbols per piece).

**Exercise 41** ($\star$)**.** Give the number of solutions in $\mathbb{N} \cup \{0\}$ for:

1. $x + y + z = 10$ where $x \geq 2$, $y \geq 1$, $z \geq 2$.
2. $x + y + z + t = 10$ where $x \geq 2$, $y \geq 1$, $t \geq 2$.
3. $x_1 + x_2 + \cdots + x_r \leq n$.
4. $x_1 + x_2 + \cdots + x_r \leq n$ where $x_1 \geq 1, x_2 \geq 2, \cdots, x_r \geq r$.

**Exercise 42** ($\star$)**.** One city in Algeria has 5 clinics. Food poisoning affected 15 people. In how many ways can these poisoned people be spread over the 5 clinics, (The poisoned people are not discernible) if

1. the number of poisoned people received by each clinic is not limited?
2. a given (specified) clinic of the five receives exactly 7 poisoned people?
3. Two particular clinics A and B receive each at least 3 people?

**Exercise 43.** How many solutions does the equation $x_1 + x_2 + \cdots + x_r = n$ have, in $\mathbb{N} \cup \{0\}$, for which exactly $k$ variables are zero?

**Exercise 44** ($\star$)**.** Find the number of integer solutions to the equation $x_1+x_2+x_3+x_4+x_5+x_6 = 60$, in each of the following cases:

(i) $x_i \geq i-1$ for each $i = 1, 2, \cdots, 6$

(ii) $x_1 \geq 2, x_2 \geq 5, 2 \leq x_3 \leq 7, x_4 \geq 1, x_5 \geq 3$ and $x_6 \geq 2$.

**Exercise 45.** Find the number of integer solutions to the equation $x_1 + x_2 + x_3 + x_4 = 30$ in each of the following cases

(i) $x_i \geq 0$ for each $i = 1, 2, 3, 4$;

(ii) $2 \leq x_1 \leq 7$ and $x_i \geq 0$ for each $i = 2, 3, 4$.

(iii) $x_1 \geq -5$, $x_2 \geq -1$, $x_3 \geq 1$ and $x_4 \geq 2$.

(iv) $x_i$ is positive and odd for each $i = 1, 2, 3, 4$.

**Exercise 46** ($\star$)**.** An elevator leaves the ground floor with 8 athletes (5 Algerians and 3 Tunisians). When it reaches the $6^{th}$ floor, it is empty. In how many ways can the athletes leave the elevator at floors 1 to 6 if: no distinction is made between the athletes (we only record how many leave at each floor)? If we distinguish them only by their nationality?

**Exercise 47.** Twenty different books are to be put on five distinct book shelves, each of which holds at least twenty books.

1. How many different arrangements are there if you only care about the **number** of books on each shelf? (Assume the books are indistinguishable for this part).
2. How many different arrangements are there if you care about which books are on which shelf, but the order of the books on their individual shelves doesn't matter?
3. How many different arrangements are there if the order of the books on each shelf does matter?

**Exercise 48.** How many integer solutions exist for the equation

$$(x_1 + x_2 + \cdots + x_m)(y_1 + y_2 + \cdots + y_n) = 25,$$

given that $x_i$ are all of the same sign, and $y_i$ are all of the same sign (zero is allowed)?

**Exercise 49** ($\star$)**.** Let $p_1 < p_2 < \cdots < p_k$ be the first $k$ prime numbers, with $k \geq 10$, and let $N \geq 1$. We form positive integers $m$ as products of powers of some of these primes, where every prime that appears in $m$ has an exponent between 1 and $N$. How many such integers $m$ can be formed in each case?

1. $m = p_1^{n_1} \cdots p_{10}^{n_{10}}$ uses exactly the first 10 primes.
2. $m$ uses exactly 10 distinct primes chosen among $p_1, \ldots, p_k$.
3. $m = p_1^{n_1} \cdots p_k^{n_k}$ uses all the first $k$ primes, and moreover (independently):
   (a) the exponents $n_1, \ldots, n_k$ are pairwise distinct (assume $k \leq N$);

(b) $n_1 + n_2 + \cdots + n_k \leq N$;

(c) the parities of $n_1, n_2, \ldots, n_k$ alternate; treat the cases $N$ even, and $N$ odd with $k$ odd.

**Exercise 50.** 1. Let $f$ and $g$ be two functions of class $\mathcal{C}^\infty$. Establish Leibniz's formula for the $n$-th derivative of a product by induction:

$$(f \cdot g)^{(n)} = \sum_{k=0}^{n} \binom{n}{k} f^{(k)} g^{(n-k)}$$

2. Show by induction or otherwise that for all $x \in ]-1, 1[$ and integer $p \geq 0$,

$$\frac{1}{(1-x)^{p+1}} = \sum_{n=p}^{\infty} \binom{n}{p} x^{n-p} = \sum_{k=0}^{\infty} \binom{k+p}{p} x^k$$

**Exercise 51** (homework)**.** 1. **Binomial Inversion:** Let $(f_n)_{n\geq 0}$ and $(g_n)_{n\geq 0}$ be two sequences of real numbers such that, for any integer $n \geq 0$,

$$f_n = \sum_{k=0}^{n} \binom{n}{k} g_k$$

Show that, for any integer $n \geq 0$, the sequences are related by the inversion formula:

$$g_n = \sum_{k=0}^{n} (-1)^{n-k} \binom{n}{k} f_k$$

2. **Surjections:** Let $E = \{1, 2, \cdots, p\}$ and $F = \{1, 2, \cdots, q\}$ with $p \geq 1$. Let $S_{p,q}$ be the number of surjections from $E$ to $F$.

(a) What is the number of functions from $E$ to $F$, whose image is $I_k$, for a fixed subset $I_k \subseteq F$ with $\mathrm{Card}(I_k) = k$?

(b) What is the number of functions from $E$ to $F$, whose image has cardinal $k$?

(c) By classifying all $q^p$ functions from $E$ to $F$ by the size of their image, show:

$$q^p = \sum_{k=1}^{q} \binom{q}{k} S_{p,k}$$

(d) Using the inversion formula from part 1, deduce the formula for $S_{p,q}$:

$$S_{p,q} = \sum_{k=1}^{q} (-1)^{q-k} \binom{q}{k} k^p$$

3. **Derangements:** A derangement is a permutation without fixed points. Let $D_n$ be the number of derangements of $n$ elements. Let $P_n^k$ be the number of permutations of $\{1, \cdots, n\}$ having exactly $k$ fixed points.

(a) Show that $P_n^k = \binom{n}{k} D_{n-k}$, with the convention $D_0 = 1$.

(b) Show that $n! = \sum_{k=0}^{n} P_n^k$.

(c) Using inversion, establish that

$$D_n = n! \sum_{p=0}^{n} \frac{(-1)^p}{p!}$$

and deduce from this the expression for $P_n^k$:

$$P_n^k = \frac{n!}{k!} \sum_{p=0}^{n-k} \frac{(-1)^p}{p!}$$

**Exercise 52** (homework)**.** Let $S(p, n)$ be the number of ways to partition a set of $p$ distinguishable balls into $n$ indistinguishable, non-empty boxes.

1. By considering the $p$-th ball, establish the recurrence relation:

$$S(p, n) = S(p-1, n-1) + n \cdot S(p-1, n) \quad \text{for } p \geq n \geq 1$$

(with the conventions $S(0, 0) = 1$, $S(p, 0) = 0$ for $p \geq 1$, and $S(p, n) = 0$ for $n > p$).

2. Calculate:

(a) $S(p, 1)$ and $S(n, n)$

(b) $S(p, 2)$ and $S(p, 3)$

(c) $S(p, p-1)$, $S(p, p-2)$, and $S(p, p-3)$